

Temporal Trends in Secondary Cementation of Rhodolith Protuberances from Army Bay, New Zealand

RESEARCH PROJECT IMPLEMENTATION

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Abstract

Rhodoliths are important marine carbonate producers that provide habitat for diverse benthic communities. Secondary cementation contributes to their structural stability and carbonate budget, but temporal patterns of cement formation is poorly understood. This study investigated changes on cementation intensity over time in six rhodolith samples from Army Bay, New Zealand. Microscopy images were analysed using ImageJ to quantify cemented areas and calculate annual cement ratios. Statistical analyses included Shapiro-Wilk, Kruskal-Wallis, pairwise Wilcoxon tests and linear regression models. Significant differences in cement ratio were found among samples, and several samples exhibited a significant decline in cementation through time. Overall, the results indicate a long-term decrease in cementation intensity, highlighting the need for further research into the environmental controls and ecological implications of this process.

1. Introduction:

1.1 CCA and Rhodoliths

Coralline algae (CCA) are a phylogenetic group of heavily calcifying macro algae (Teichert 2024), they occur globally in recent environments over the marine photic zone, ranging from tropic to polar latitudes (Teichert et al. 2012; van der Heijden und Kamenos 2015; López Correa et al. 2023), and also across a range of bathymetric settings, from deeper waters up to 268 meters in depth, known as deepest plant life on Earth (Littler et al. 1985), to shallow reef crests (Dean et al. 2015). As a reaction to different environmental settings, and because of their high phenotypic plasticity (Teichert 2024; Pigliucci et al. 2006)) CCA can develop different thallus morphologies within the same species (Fusco und Minelli 2010). They develop in two different ways, the first being as a thin attached crust to a fixed substrate and the second being as free living rhodolith (Teichert 2024). Rhodoliths act as ecosystem engineers by crating complex three-dimensional habitats (Teichert 2024). They can form either through spore settlement on hard substrate or by encrusting portions of detached protuberances. Depending on the developmental form they are either made completely out of CCA or can include little rock or shell fragments as their core. (WJ Woelkerling 1988; Foster 2001a). Rhodoliths are subject to a range of biotic interactions that significantly influence their morphological development and ecological role. These interactions can be categorized into constructive and destructive processes, each contributing differently to rhodolith growth and modification. Rhodoliths serve as a stable substrate for an oligospecific assemblage of epiphytic organisms, including encrusting brown algae, tube worms, vermetids, bryozoans, barnacles, gastropods and chitons (Foster 2001). These organisms contribute to rhodolith growth as their skeletal remains are overgrown by coralline algae, enhancing structural complexity. Destructive processes involve mostly bioerosion by various organisms producing borings, ranging from microborings from 0,5-1mm to large borings of 8-10 mm produced by polychaete worms and bivalves (Basso et al. 2009). Growth rates of corallines are generally very slow, ranging from 10 (Littler et al. 1991) to 5200 $\mu\text{m yr}^{-1}$ (Adey und Vassar 1975), while the average is about 500 $\mu\text{m yr}^{-1}$ (Schafer et al. 2011). An important feature resulting from both cell growth and changing seasonal environmental conditions is the formation of growth increments. These increments commonly appear as banding patterns starting from the core of the coralline, similar to the age rings in trees (Teichert 2024). Most studies interpret growth increments in CCA as forming on annual scale, where lighter bands with increased growth are correlated with summer months, whereas

the darker bands appear to result from reduced calcification during the darker winter months (Freiwald 1993).

1.2 Cell structure and Cementation

Their mineralized skeleton is made of high-Mg- calcite, usually with an Mg-content of 7-20 mole percent MgCO_3 (Chave 1954; Moberly 1970; Milliman et al. 1971), but also contain cell and pore infill by more stable carbonates, aragonite and dolomite (Díaz-Pulido et al. 2014). The morphology of the CCA tissue is differentiated into three parts, hypothallium, perithallium and epithallium (Johnson 1962). The Hypothallium is the basal part of the crust with elongated cells, that initially grow horizontally and later curve upward. The perithallium is composed of vertical threads of cells and the calcite walls are slightly thicker than in the basal section. The epithallium cells are not calcified and therefore not visible in the skeleton (Alexandersson 1977).

A key feature in CCA is non-skeletal cementation, which has been partially studied. There are different approaches trying to explain the cause of darker cemented spots between the layered cells of the perithallium. Some authors suggest that they could be decayed starch grains (Freiwald und Henrich 1994; Alexandersson 1977). While others like (Díaz-Pulido et al. 2014) argue that cementation occurs because parts of the skeleton dissolve and then re-precipitate carbonate minerals, especially dolomite, aragonite and high-Mg-calcite within their pores and cracks. Reasons for why the skeleton dissolves could be due to increased atmospheric pCO_2 and an associated decline in seawater pH (Sandberg 1983), but also mechanical triggers through grazers like Annelids, Arthropods, Mollusks, Echinoids and fishes that coexist in Rhodolith environments and use their surfaces as food source, while feeding on microbial films or sometimes the algal tissue itself (Steneck, 1983).

1.3 Aim of the Study

A range of morphological and taphonomic parameters have been described for CCA and rhodoliths, particularly in relation to their growth patterns, ecological roles and preservation potential. Many studies have examined the ultrastructure of the algae focusing on cell size, growth bands and their potential to reflect seasonal or climatic variability. These bands have been used as indicators of environmental conditions such as temperature shifts, light availability and even repositioning of the coralline, thus carrying very important information (Foster 2001). While cementation in corallines is mentioned in the literature to some extent, little research has directly investigated its drivers or its capacity to store carbon, leaving a huge gap in understanding rhodolith ecology. Given that rhodolith beds are major CaCO_3 bio factories (Amado-Filho et al., 2012), clarifying their cementation processes is essential for evaluating their role in the marine carbon cycle and strengthening arguments for their protection as carbon-rich habitats in the context of blue carbon research (Amado-Filho et al., 2012).

The aim of this study is to investigate whether the internal cellular structures of rhodoliths from the Whangaparaoa Peninsula exhibit non-skeletal cementation within their growth bands. By examining banding patterns as years and cementation size distribution within them, this research explores the extent to which cementation is recorded in the CCA. It is expected that between multiple samples a trend towards increased cementation through time could be found preserved in the rhodolith structure.

2. Material and Methods

2.1 Sampling site

The Rhodolith protuberances examined in this study were collected at Army Bay ($36^{\circ}35.92'S$, $174^{\circ}48.91'E$) on the northern coast of the Whangaparaoa Peninsula, North Island of New Zealand (**Figure 1**). This site is located within the western Hauraki Gulf north of Auckland, which is sheltered from the Southern Ocean (Basso et al. 2009). The Gulf has a humid oceanic climate. The tidal ranges show significant local variations, with a mean tidal range at 2.18m. (Walters et al., 2001). The local geology consists primarily of Miocene volcanoclastic sand sedimentary deposits of the Waitemata Group ((Ballance 1974; Nalin et al. 2008). Rhodolith formation occurs mainly within shallow rocky subtidal habitats at depths of up to approximately 5m (Basso et.al, 2009). The Rhodoliths that were chosen for this study were obtained on the 22nd of March 2023 and still alive due to its pink colouring (**Figure 2**). After air-drying, samples were stored and sent to FAU Erlangen-Nürnberg for further analysis (Teichert et al. in prep.).

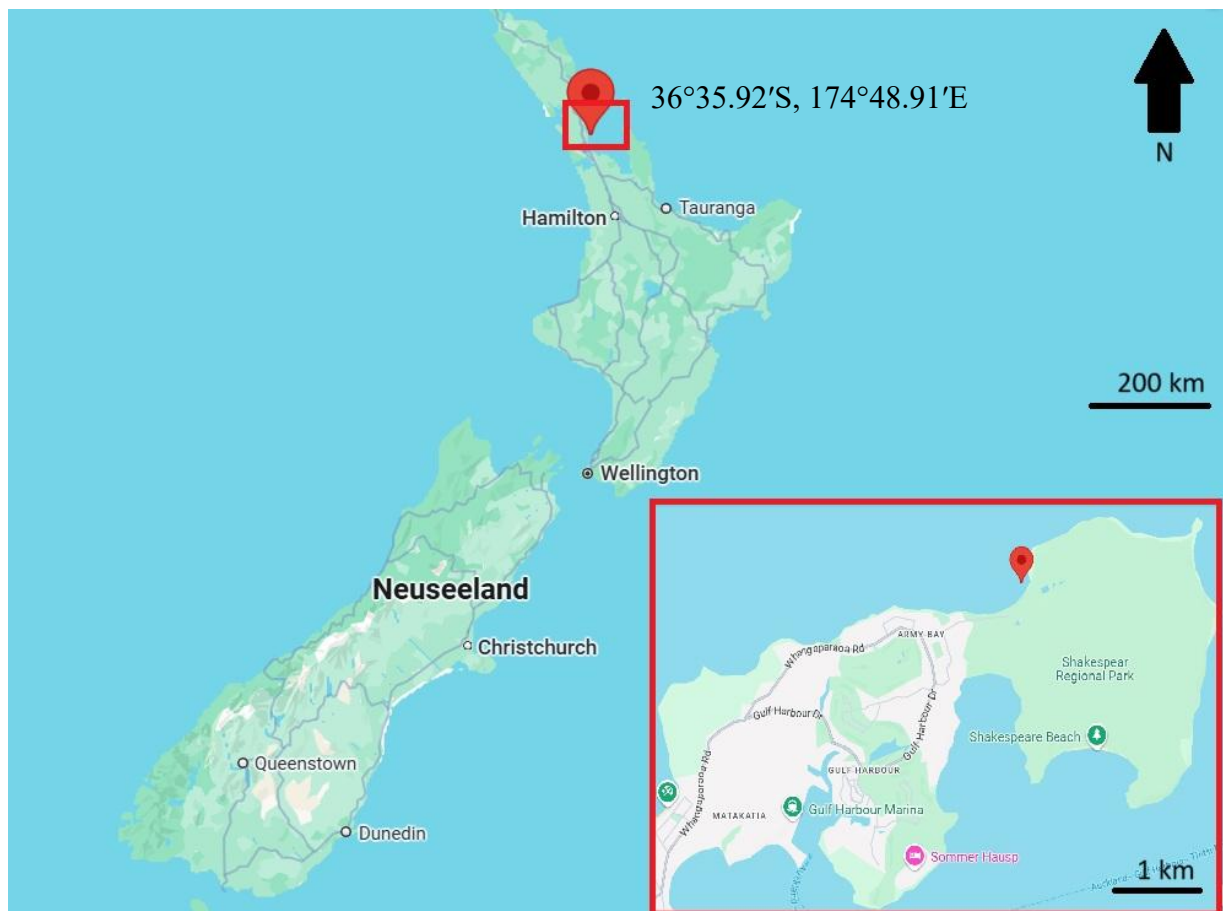


Figure 1 Rhodolith sampling site, Army Bay, Whangaparaoa Peninsula, New Zealand <https://www.google.de/maps/> (06.09.2025).

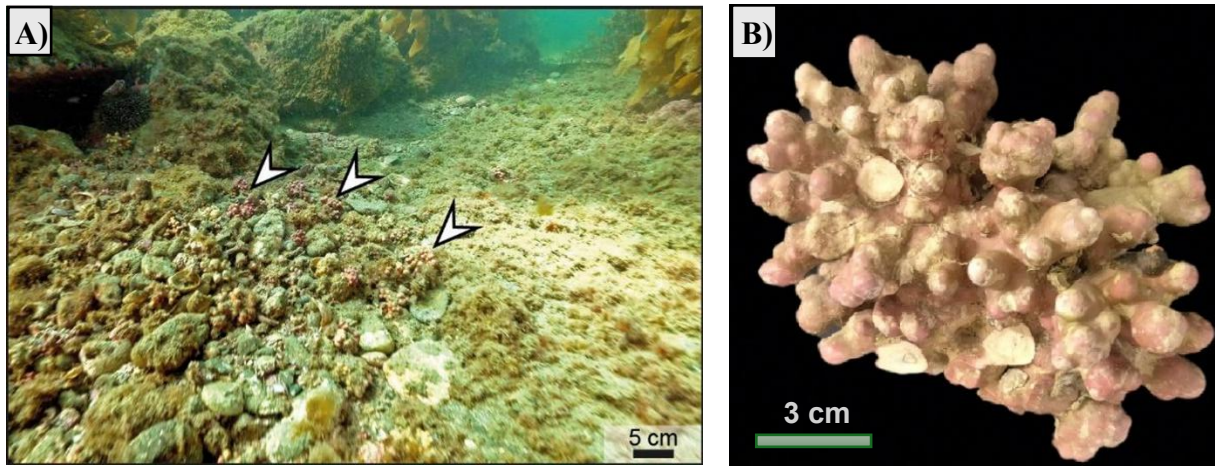


Figure 2 A) Rhodolith collection site. B) Rhodolith specimen displaying multiple protuberances, several of which are missing due to sampling for microscopy

2.2 Sample Preparation

The six samples used this study had previously been prepared for microscopic analysis. Well-preserved protuberances were removed from rhodoliths, embedded in epoxy resin, sectioned vertically using the Macatome T180 precision saw, and polished with silicon carbide 800 to ensure smooth surfaces. After airdrying for 24 hours the polished halves, they were re-embedded in resin. After another 48 hours of curing, the samples were ground on both sides with silicon carbide 800 to achieve a flat and even surface. To enhance the visibility of the cellular structures under the microscope, residual cell wall material was removed by etching the samples in 60% acetic acid for 30 seconds, rinsed with distilled water, cleaned in an ultrasonic bath. After dusting with compressed air, and drying for three days, they were sputter-coated with gold under argon gas for 6 minutes using 2-minute intervals with 30-second pauses at 40 mA and 10,000 V. Polished sections were imaged using Zeiss AXIO Zoom. V16 microscope in reflected-light mode. High-resolution composite images of each protuberance were generated by tile acquisition with 10% overlap between adjacent images. The individual tiles were subsequently stitched together to create complete composite images of the vertical section. To improve visibility and contrast of growth increments, the colour images were converted to grayscale using Adobe Photoshop 2022. The resulting black-and-white images enhanced the distinction between annual growth bands and facilitated further image analysis. The images were examined to assess growth band patterns, cellular preservation, and cementation features along the longitudinal axis of each protuberance.

2.3 Data Collection and Statistical analyses

Image analysis was carried out in ImageJ using a custom macro. For each sample, the image scale was calibrated and a standardized rectangular region of interest with a width of 1 mm was positioned along the growth axis of the protuberance to ensure comparability among samples. Annual growth increments were identified based on alternating banding patterns visible in the algal tissue and used to assign individual years. Areas of visible secondary cementation were then manually delineated based on their darker appearance in the images, and their surface area was measured within each annual increment (**Figure 3**). This procedure resulted in the identification of annual growth bands and the quantification of cemented areas for each sample.

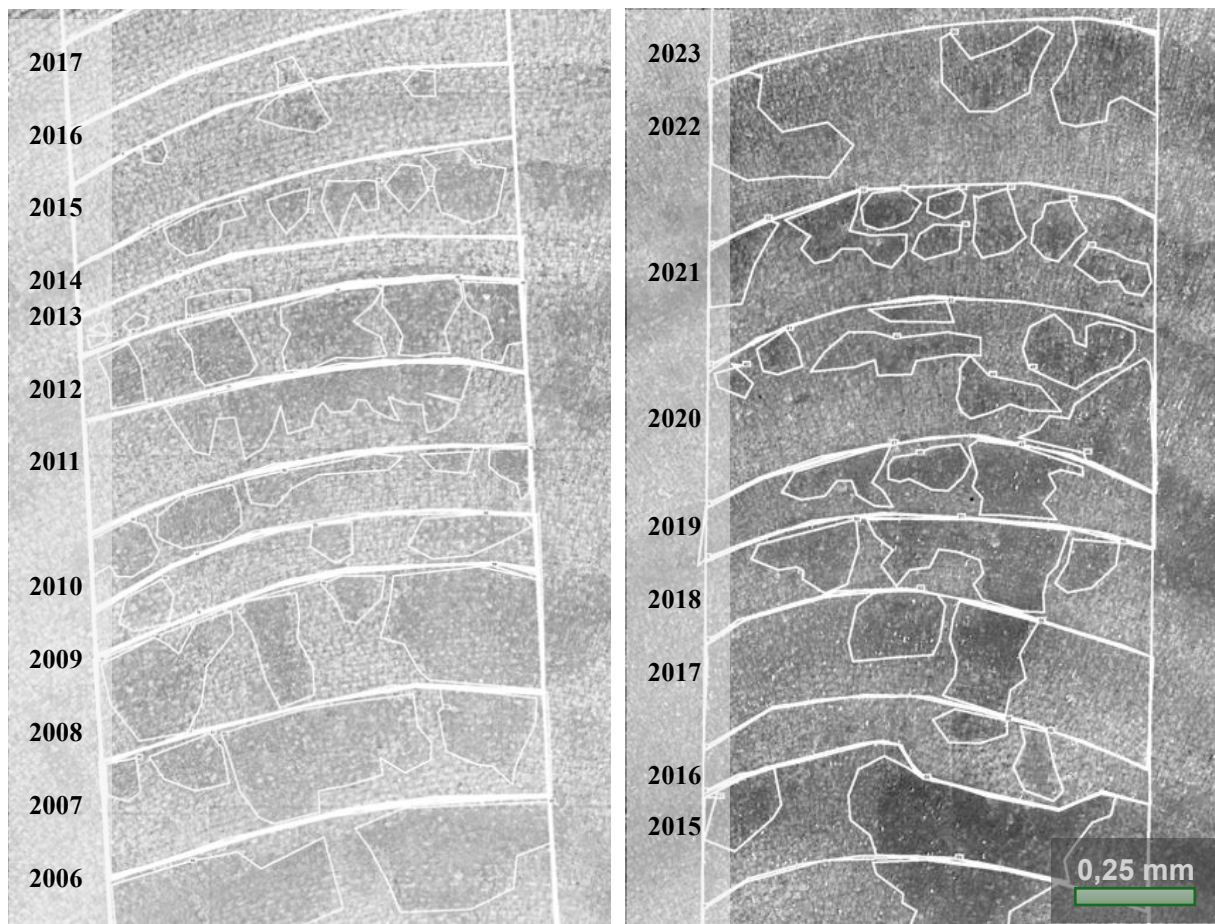


Figure 3 representative images of two rhodolith protuberances used for cementation analysis. White lines indicate annual growth increments, and outlined polygons represent areas identified as secondary cementation. A) Sample 120 spanning the years 2006-2017 B) Sample spanning the years 2015-2023.

After image analysis, the cementation measurements were assigned to their years using Microsoft Excel spreadsheets. All subsequent statistical analyses were performed in RStudio (Version 4.5.3). For each sample, the cement ratios were calculated as the area of total cement area to total area of the assigned year. Prior to statistical comparisons, the cement ratio data was assessed using the Shapiro-Wilk test for normality. As several datasets did not meet the assumptions of normality, non-parametric statistical tests were applied. Differences in cement ratio among samples were evaluated using a Kruskal-Wallis rank-sum test. When significant differences were detected, pairwise Wilcoxon rank-sum tests were performed. Further Temporal trends in cement ratio were investigated using linear regression models for each sample individually, with year as the predictor variable. For each sample, the regression slope, adjusted R^2 , and p-value were recorded. In addition, a combined linear model including year, sample identity and their interaction was used to test whether temporal trends differed among samples. To assess the overall temporal trend independent of the most distinct sample, an additional linear regression analysis was performed on the annual mean cement ratio calculated from all samples excluding sample 145. Graphical visualisations, including regression plots and comparative figures, were generated using the ggplot2 package in R.

3. Results

At first the normality was assessed using the Shapiro-Wilk test. It revealed that the cement ratio was not normally distributed in four of the six analysed samples. Samples 100, 106, 120 and 123 had p-Values of $p < 0.01$ and sample 141, $p = 0.9389$ and 145, $p = 0.083$. They did not significantly deviate from a normal distribution. Given the predominance of non-normal distributions, non-parametric statistical methods were applied for further comparisons. The Kruskal-Wallis test indicated significant differences in cement ratio among the samples (chi-squared = 77.938, $df = 5$, $p\text{-value} = 2.265e-15$). The Wilcoxon test demonstrated that samples 141 and 145 differed significantly from several other samples. While Sample 145 showed significant differences from all other samples (all $p < 0.001$), sample 141 differed significantly from all other samples but sample 145. Among the samples 100, 106, 120 and 123 no significant differences were detected. Linear regression analyses revealed temporal trends within individual samples (**Figure 4**). There are significant negative relationship between year and cement ratio in samples 100, 120 and 123. The estimated slopes were (-0.011820 , Adj. $R^2 = 0.2722$, $p < 0.001$) for sample 100, (-0.016811 , Adj. $R^2 = 0.2174$, $p = 0.007$) for sample 120, and

-0.013285, Adj. $R^2 = 0.171$, $p = 0.02256$) for sample 123. For the other samples no significant temporal trend was observed, all p -values $p > 0.05$.

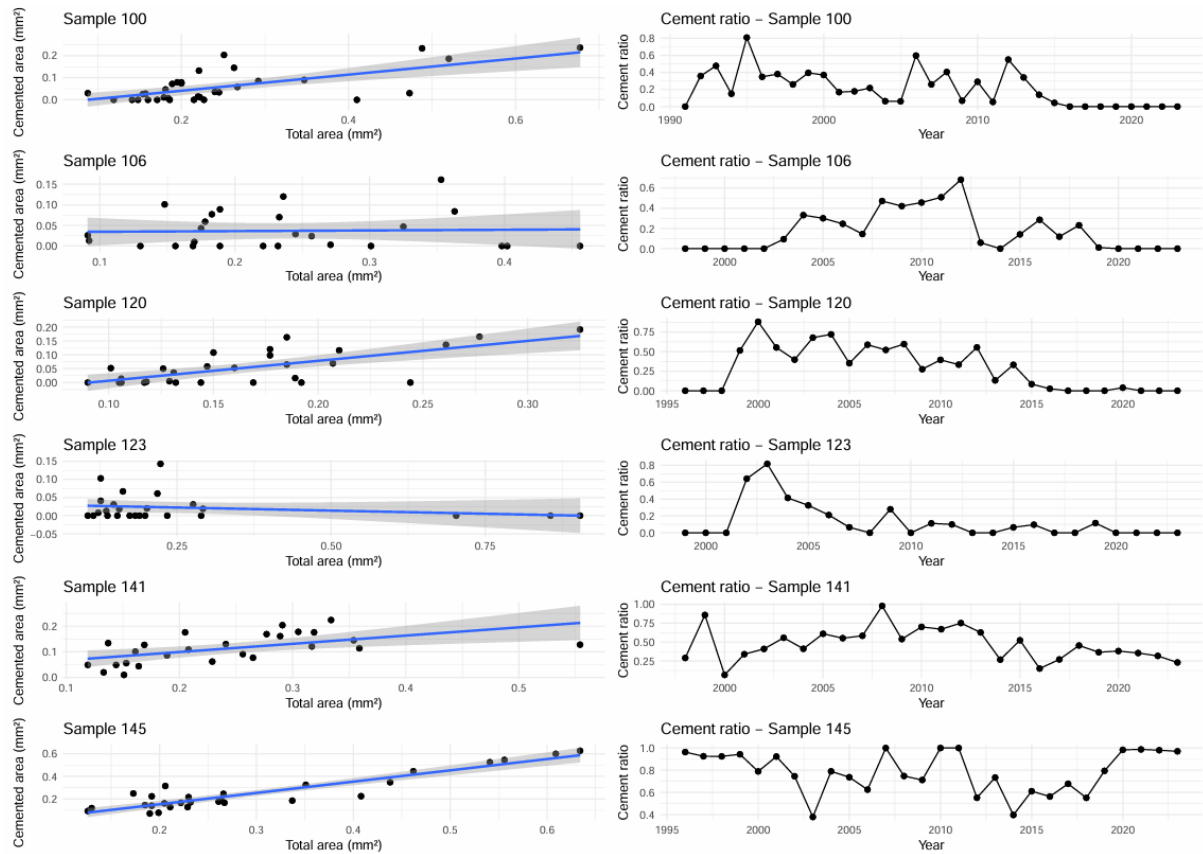


Figure 4 Relationships between total measured area and cemented area (left column) and temporal variation in cement ratio (right column) for Samples 100, 120, 123, 141, and 145. Blue lines indicate fitted linear regression with 95% confidence intervals. Temporal plots show annual cement ratio values calculated from the proportion of cemented area relative to total skeletal area.

A combined linear model (**Figure 5**) was used to assess the effects of year and sample identity on cement ratio. The model revealed significant effects of both year and sample identity (both $p < 0.001$). However, the interaction between year and sample was not significant ($p = 0.233$).

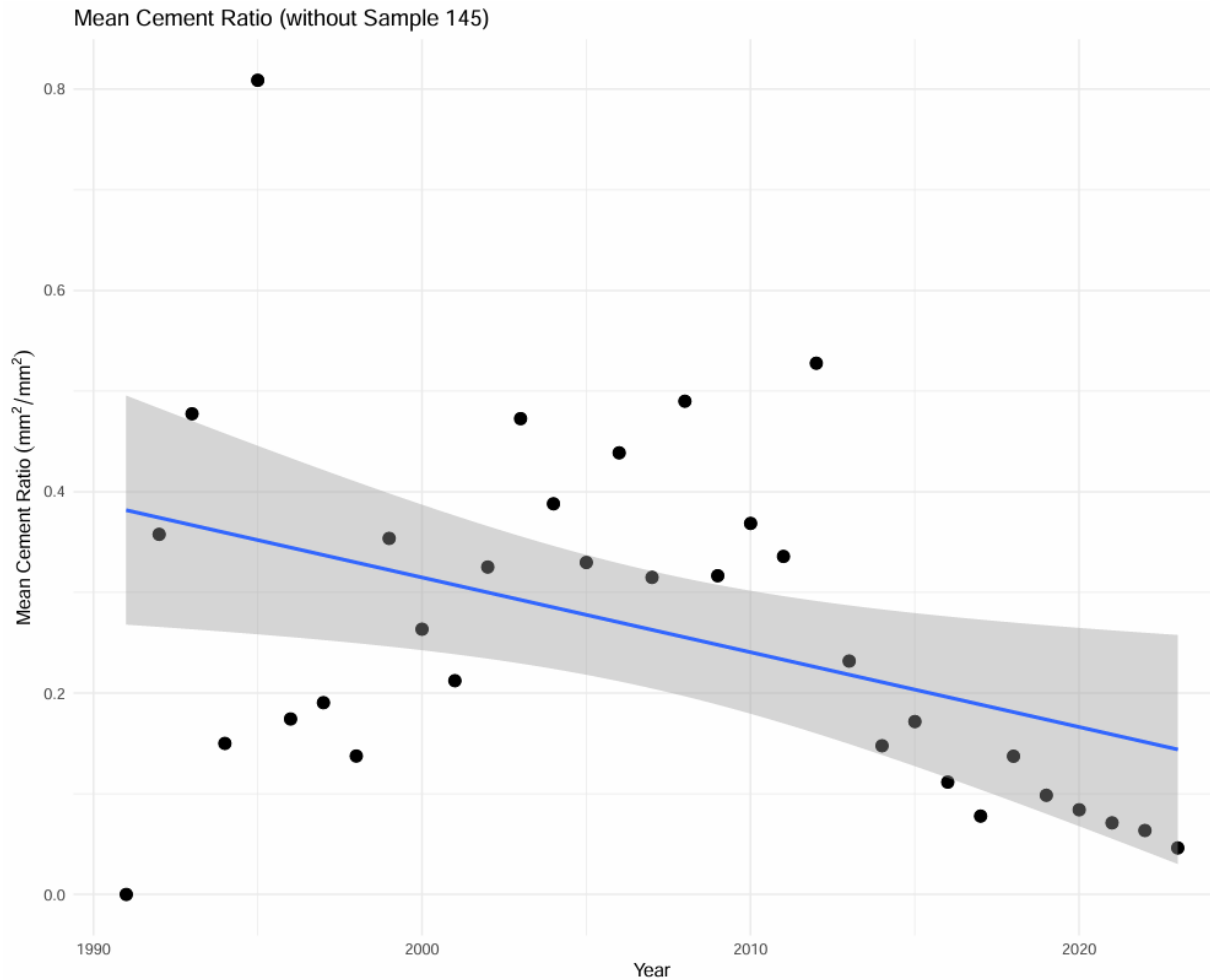


Figure 5 Mean cement ratio through time calculated from Samples 100, 106, 120, 123 and 141, excluding Sample 145. Points represent annual mean cement ratios. While blue line shows fitted linear regression and the grey area shows the 95% confidence interval.

4. Discussion

The aim of this study was to investigate temporal patterns of secondary cementation in rhodolith protuberances and to determine whether cementation intensity has changed over time. The results revealed significant differences in cement ratio among samples and identified a significant overall decline in cementation through time. The Kruskal-Wallis and pairwise Wilcoxon tests demonstrated that cement ratios differ significantly among several samples. In particular sample 141 and 145 exhibited significantly higher cement ratios compared to the other samples. These differences are likely to reflect local variability in growth history, microenvironmental conditions or biological characteristics of individual rhodoliths (Foster 2001). In sample 145, the algal tissue was already dead at the time of collection, which is consistent with the stronger cementation observed. (Díaz-Pulido et al. 2014) show that dead

coralline tissue is highly susceptible to dissolution, especially in the unpigmented skeleton, and that dissolution promotes secondary precipitation of dolomite and aragonite within pores. The exposed, unprotected surface of the dead portion of sample 145 therefore likely underwent enhanced chemical alteration. Individual linear regression showed significant negative temporal trends in Samples 100, 120 and 123, while Samples 106, 141 and 145 showed no statistically significant trends. However, the combined model excluding Sample 145 both revealed an overall decline in cement ratio through time. This suggests that the observed decrease is not driven by a single sample but represents a broader pattern present across multiple rhodoliths, suggesting that a common process like ocean acidification, warming or enhanced biotic interactions (Wilson et al. 2004) may influence cementation across the study site.

Several limitations should be considered when interpreting these results. First the number of available samples was limited, reducing the statistical power of the analyses. Second, measurements of annual increments and cemented areas were performed manually in ImageJ and therefore may be affected by observer bias. Although a standardized workflow was applied to all samples, small differences in delineation of annual bands or cemented regions can not be excluded. Third, most of the data were not normally distributed, requiring the use of non-parametric statistical methods. While these methods are robust, they generally provide less detailed information than parametric approaches. Furthermore, cementation, and growth of CCA is likely influenced by multiple environmental biological factors that were not included in this study. Variables such as seawater temperature, carbonate chemistry, nutrient availability, hydrodynamic conditions (Halfar et al. 2000), and growth rates may all affect the formation of secondary cements (Alexandersson 1977).

Consequently, year alone is unlikely to explain the full variability observed in the cement ratios, as also indicated by the relatively low to moderate R^2 values of the regression models. Despite these limitations, this study contributes to closing an important knowledge gap regarding temporal variability of cementation in rhodoliths. While previous studies have documented the occurrence of secondary cementation in CCA, quantitative investigations of long-term changes in cementation intensity are scarce (Amado-Filho et al. 2012). This study provides evidence, that cementation can vary significantly through time and established a framework for future analyses of environmental controls on this process

5. Conclusion

This study investigated temporal patterns of secondary cementation in rhodolith protuberances from New Zealand. Significant differences in cement ratio were observed among samples, and the overall analyses revealed a significant decline in cementation intensity through time. Although the underlying causes remain unclear, the results suggest that this trend may represent a broader pattern. As secondary cementation is a common feature of rhodoliths {...} and may influence their structural stability and preservation, further research is needed to identify the environmental factors controlling this process.

A better understanding of these mechanisms will contribute to the conservation and management of rhodolith ecosystems under changing environmental conditions.

6. Acknowledgements

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7. Supplementary material

The supplementary material associated with this study, including datasets, ImageJ macros and R scripts used for data processing and statistical analyses, is available on GitHub at: [marialuisaerlangen/RPI: Temporal Trend Analyses in Secondary Cementation of Rhodolith Protuberances from Army Bay, New Zealand](https://github.com/marialuisaerlangen/RPI: Temporal Trend Analyses in Secondary Cementation of Rhodolith Protuberances from Army Bay, New Zealand)

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